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Comparing the effect of mindful and other engagement interventions in nature on attention restoration, nature connection, and mood☆

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ABSTRACT

Mindful engagement interventions have been shown to improve psychological benefits of nature experiences but to date there has been little evaluation of their efficacy compared with other forms of engagement. An online experimental study was conducted to compare different forms of engagement with nature. Before and after a 20-min outdoor experience, participants ($n = 215$) completed surveys on state-mindfulness, connection with nature, and mood, and then the Sustained Attention to Response Task (SART). Participants were randomly allocated to one of four engagement intervention groups: mindful engagement, directed engagement, mind wandering, and an unguided control group. The groups did not differ on connection with nature or positive affect after the nature experience, when accounting for the baseline measures, however the mind wandering group reported significantly lower negative affect than the directed engagement group. Overall, SART outcomes indicated that the unguided control group had the greatest level of attention restoration, suggesting that all three engagement interventions taxed the attention system. Mediation analyses indicated that there was an indirect effect of state mindfulness on the psychological outcomes, primarily for the mindful engagement group when contrasted with the unguided control. The findings challenge the idea that all forms of heightened engagement enhance the experience of nature and resulting psychological outcomes. We use these findings to outline considerations for designing and applying engagement interventions in nature for specific outcomes.

1. Introduction

Research over the past decade has explored the effect of various nature-based engagement interventions on psychological benefits of spending time in nature (Duvall, 2011; Korpela et al., 2017; Lumber et al., 2017). Several of these interventions apply qualities of mindfulness to the nature experience, such as nonreactivity and curiosity (e.g., Lymeus et al., 2018; Nisbet et al., 2019), setting them apart from other engagement interventions including those using imagination and abstract thought (e.g., Duvall, 2011). Concurrently, researchers have hypothesised and observed varying psychological outcomes associated with engagement interventions, including mood (Korpela et al., 2017) and connection with nature (Lumber et al., 2017). Despite this diversity, only one study to date has compared outcomes of different engagement interventions on a single set of psychological outcomes (Pasanen et al., 2018). Further, theoretical accounts are emerging of mind wandering as an alternative form of engagement in nature (Williams et al., 2018), warranting its consideration alongside existing engagement concepts. To progress understanding of engagement interactions in nature, it is essential to compare and contrast mindful and other engagement interventions, so that they can be designed to support desired outcomes.

Heightened engagement in nature may influence commonly reported psychological outcomes of nature experiences. They include attention restoration (Ohly et al., 2016; Stevenson et al., 2018), mood (McMahan & Estes, 2015), and connection with nature (Lumber et al., 2017; Mayer & Frantz, 2004). Improved attention performance has been observed after two-weeks of engagement-enhancing tasks conducted in nature (Duvall, 2011), after a walking intervention with restoration-based engagement exercises (Pasanen et al., 2018), and when participants were simply instructed to 'pay attention closely' to vegetation in

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photographs (Y.-H. Lin et al., 2014). While satisfaction with engagement tasks has been associated with mood enhancement (Korpela et al., 2017), experimental studies have not found effects of engagement exercises on self-reported mood (Lumber et al., 2017; Pasanen et al., 2018). Heightened engagement with nature may also improve one's feeling of connection with nature, through a sense of *immersion* in nature (Weinstein et al., 2009), or specifically designed engagement exercises (Lumber et al., 2017). Within these studies, there is a relatively consistent finding that engagement in nature improves attention performance, with little evidence for engagement impacting mood. The mixed findings may be the result of diverse designs of engagement interventions, though associations between specific engagement forms and outcomes remain unclear. The current study aimed to experimentally examine the effect of various forms of engagement on psychological outcomes of nature. In the following section, we review development of engagement exercises in the literature, and specify theoretical contrasts between forms of engaging in nature.

1.1. Diverse engagement interventions in nature experiences

Kaplan (2001) proposed that a heightened engagement or mental resonance with the environmental experience would support restorative outcomes, and since then several studies have designed engagement tasks to strengthen outcomes of nature experiences. Duvall (2011) developed exercises to heighten interest of the environment during a two-week walking intervention. Some exercises were quite abstract (e.g., "How would this area be different if everyone had to grow their own food?"), while others were more directed (e.g., "Focus on sounds", Duvall, 2011, p. 29). The theoretical foundation for the exercises did not clearly relate to psychological restoration in Duvall's study, but other researchers have built on these ideas and designed exercises that test theoretical pathways to restorative outcomes (Korpela et al., 2017; Pasanen et al., 2018) and nature connection (Lumber et al., 2017). For example, Pasanen et al. (2018) developed restoration-enhancement exercises that conceptually align with theoretical pathways described in attention restoration theory (ART; Kaplan, 1995), and compared their efficacy with Duvall's awareness-enhancing exercises (2011). Participants in the restoration-enhancement condition had better sustained attention after the nature experience, while both the restoration- and awareness-enhancing groups reported comparable improvements in positive mood (Pasanen et al., 2018). These studies demonstrate that the form of engagement is clearly important in the effect on outcomes and thus needs to be carefully considered when designing such exercises. The current study follows this line of reasoning: we developed and contrasted theoretically driven engagement exercises, and extend past research by using mindfulness as a starting point to identify and distinguish key qualities of nature-based engagement exercises.

Mindfulness has received growing interest as a form of engagement intervention in nature experiences, and its characteristics can provide a basis to compare qualities and forms of engaging in nature (Macaulay et al., 2022). Mindfulness involves awareness of the present moment, which includes the external (sensory experience) and internal experience (thoughts and emotions), with qualities of curiosity, acceptance, and without judgment or reactivity (Bishop et al., 2006; Cardaciotto et al., 2008; Kabat-Zinn, 2003). Mindfulness thus differs from other engagement exercises that direct attention to nature: where other forms of engagement may be external (e.g., focusing on sounds and smells; Weinstein et al., 2009) or internal (e.g., observing one's mood; Korpela et al., 2017), mindful engagement can be characterised as an awareness of the whole experience – external and internal – that includes acceptance and nonjudgment (Bishop et al., 2006; Cardaciotto et al., 2008).

Mindfulness has been applied in heterogenous ways in studies of nature experience. Studies have compared the effects of established mindfulness interventions in natural settings with indoor or built environments, finding benefits of mindfulness in nature on attention performance (Lymeus et al., 2018), connection with nature, and stress reduction (Choe et al., 2020). Short walking interventions in nature with mindful engagement have demonstrated benefits for connection with nature and negative affect (Nisbet et al., 2019), and anxiety and happiness (Shin et al., 2013). Studies have found that being in nature can facilitate state mindfulness or present-centred awareness (Brymer et al., 2021), and that state mindfulness can mediate the effect of nature exposure on affect (Stewart & Haaga, 2018). Brymer et al. (2021) call for further research on the mediating effects of mindfulness in nature, to better understand its role in psychological outcomes of nature experiences. This would help delineate the key differences between mindful engagement interventions in nature and other forms of engagement in nature.

Mind wandering is a form of engagement that has received some attention for its role in supporting restoration in nature experiences (Williams et al., 2018), yet has not been applied as an engagement technique in empirical studies. Mind wandering involves an internally oriented awareness (Schooler et al., 2014), with relatively little cognitive effort or control (Christoff et al., 2016). While mind wandering is commonly linked with negative outcomes such as lower mood (Killingsworth & Gilbert, 2010), it may lead to improvements in mood when the content of thoughts is interesting or stimulating (Franklin et al., 2013; Schooler et al., 2014). Mind wandering in nature has also been proposed to support attention restoration via a process of gentle shifting between internal mind wandering and external fascination with nature (Williams et al., 2018). There is some evidence for these ideas, where settings that are highly rated for restorativeness are also rated highly for the potential for introspection and mind wandering (Cervinka et al., 2020). Mind wandering offers an interesting research direction in the context of engagement in nature experiences. However, with mixed findings on the potential for mind wandering to improve or lower mood (Franklin et al., 2013; Killingsworth & Gilbert, 2010), we would expect exercises that are designed to draw awareness to nature (e.g., through mindfulness or other awareness-enhancing activities) would be more restorative than internally-oriented mind wandering.

1.2. The current study

This study aimed to compare the effectiveness of three different engagement interventions in nature, on psychological outcomes. During a brief nature experience, a *mindful* engagement, a *directed* engagement, and a *mind wandering* intervention were compared with an *unguided* control condition. The interventions were designed to be theoretically distinct from one another. The mindful engagement intervention was designed with reference to prior studies that implement short, guided mindfulness interventions focusing on the breath and physical sensations (Arch & Craske, 2006; Aspy & Proeve, 2017) and to internal sensations (Kiken & Shook, 2011), and studies that apply mindfulness in nature (Lymeus et al., 2018; Nisbet et al., 2019). The directed engagement intervention was designed by drawing on exercises shown to be effective in encouraging attention to the nature experience (e.g., Duvall, 2011), but with qualities that contrast with mindfulness (for example, encouraging elaborative and judgmental thought). Additionally we drew on designs of active control groups in mindfulness literature, that use physical relaxation exercises (Aspy & Proeve, 2017; Rahl et al., 2017), movement and imagery (MacCoon et al., 2012). The mind wandering intervention was based on studies that use mind wandering

as a control condition for mindfulness interventions, where participants are instructed to let attention move away from the present moment (Arch & Craske, 2006; Kiken & Shook, 2011).

To examine whether mindfulness itself is an important factor in such interventions, we additionally aimed to test whether the level of state mindfulness during the experience played a key role. We aimed to contribute to existing literature that explored the role of state mindfulness during nature exposure (e.g., Brymer et al., 2021; Stewart & Haaga, 2018) by testing whether state mindfulness mediated the effects of the engagement interventions on psychological outcomes of the nature experience.

Hypothesis 1. A mindful engagement intervention would improve outcomes of, a) state mindfulness, b) connection with nature, c) positive affect, d) negative affect, and e) sustained attention, to a greater degree than other intervention groups.

Hypothesis 2. The directed engagement intervention would improve all outcomes greater than the mind-wandering intervention and the unguided control group.

Hypothesis 3. Engagement interventions in nature would be indirectly associated with psychological outcomes through a higher level of state mindfulness during the experience.

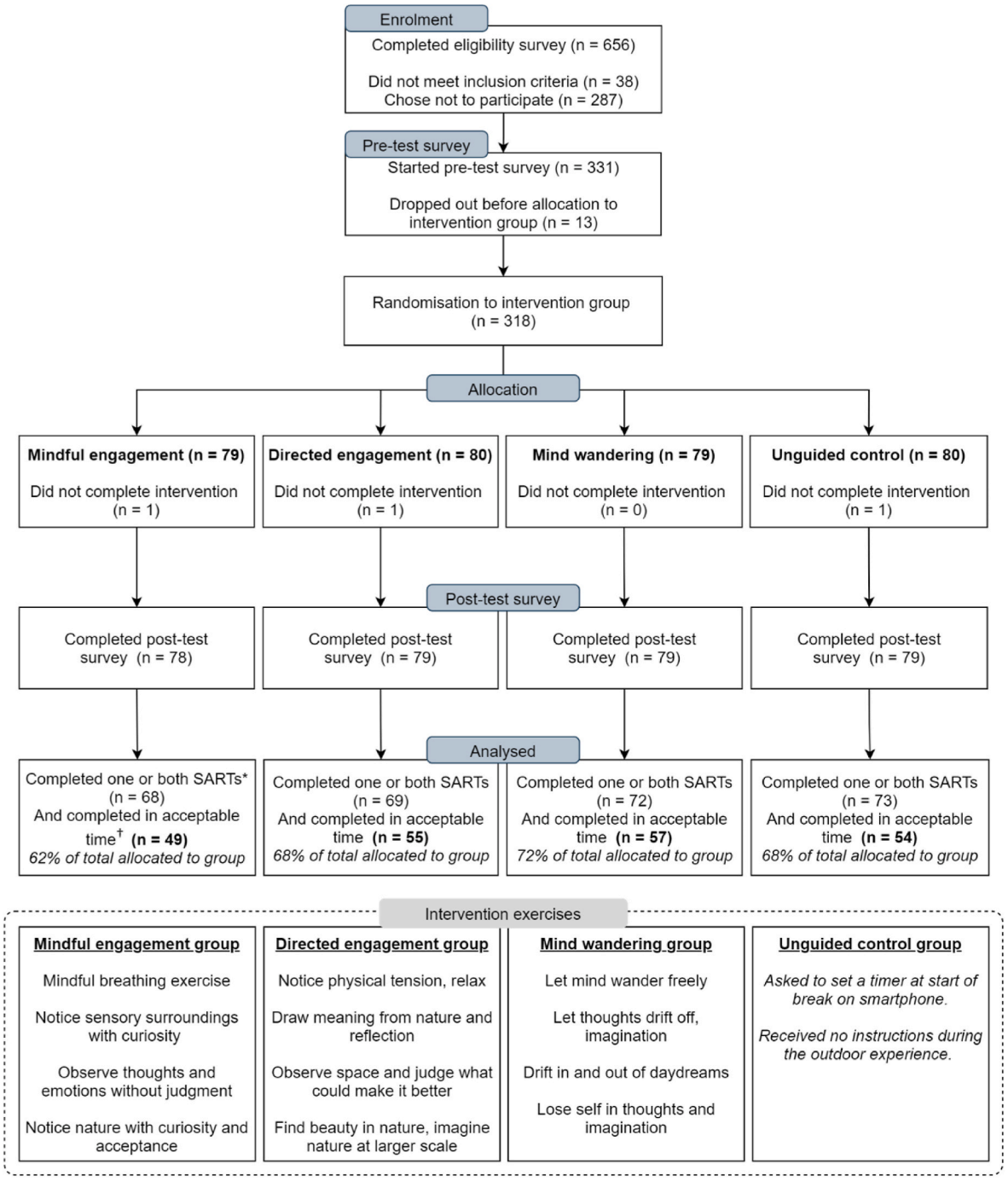


Fig. 1. Flowchart of participant enrolment, completion rates, and overview of intervention exercises.
*Participants who completed one or both the pre- and post-test SARTs were included in the final analysis for other outcome measures.
†Participants 'completed in acceptable time' if the time spent during the intervention was between 20 and 45mins.

2. Method

2.1. Design

Surveys and an attention task were completed online before and after a 20-min outdoor experience. Participants were randomly allocated to one of four intervention groups that engaged in different ways during their outdoor experience: mindful engagement, directed engagement, mind wandering, and an unguided control group. The study design and hypotheses 1 and 2 were preregistered, and the study was approved by the Faculty of Science Human Ethics Advisory Group, University of Melbourne (ID: 2057374.2).

2.2. Participants

Minimum sample size was determined with a power analysis that considered several factors. Prior research that tested group differences on engagement in nature demonstrated reasonably large effects on nature connection (Lumber et al., 2017) and attention (Y.-H. Lin et al., 2014). Our intervention groups were designed to have some conceptual overlap, so we anticipated smaller effects between groups and used a more conservative effect size in a GPower analysis (Faul et al., 2007) with the following specifications: $\alpha = .05$, power level = .80, and $f = .25$, sample size = 175. Fritz and Mackinnon (2007) suggest a sample size of 148 will power a mediation analysis with small-medium effect sizes on both *a* and *b* pathways. Considering these analyses, and expected attrition rates on online surveys, we aimed to recruit a minimum of 250 participants and retain at least 175 in final analyses.

Participants were recruited using online advertising (staff and student newsletters). Psychology undergraduate students were eligible for 1% credit to their grades for participating. Other participants were eligible to go in the draw to win a gift voucher. Inclusion criteria were age (18 years or older), currently working or studying on some days, access to an outdoor space to take a break, access to a computer and a smartphone. The final sample of participants that completed the pre- and post-test measures was 215 (Fig. 1). Demographic characteristics of the total sample and each intervention group are listed in Table 1. The four intervention groups did not differ on demographic variables.

2.3. Procedure

The study was conducted entirely remotely, between October 2020 to March 2021. Participants received a survey link and were instructed to participate during their work or study day and treat the outdoor experience as a work break. The survey began with information about the study – without specific information about the experimental conditions – and informed consent to proceed. The pre-test survey followed, and contained a series of demographic questions, measures of trait connection with nature, trait mindfulness, current mood, and the Sustained Attention to Response Task (SART). The pre-test SART was expected to induce attentional fatigue to increase the potential for restorative effects during the nature experience (Kaplan, 1995).

Immediately after the SART, participants were instructed to go to an outdoor space near their workplace (at home, if they were working from home), and access the survey on their smartphone. The survey software randomly allocated participants to groups at this point. The intervention for all participants was to spend 20 min in an outdoor environment. This time frame is suitable for the context of a work break (Brown et al., 2012), and has been shown in similar studies to lead to restorative outcomes (Lumber et al., 2017; Nisbet et al., 2019). Participants in the experimental conditions (mindful engagement, directed engagement, and mind wandering) received short text instructions on their smartphone, at 5-min intervals, about how to engage during the experience. The unguided control group was instructed to set a timer and finish their break after 20 min. No further instructions were given to participants on what to do during this break. An overview of intervention exercises is presented in Fig. 1 and full instructions are detailed in Appendix A.

After the outdoor break, participants were instructed to commence the post-test measures on their computer. The post-test began with the SART, followed by measures of current mood, state mindfulness and mind wandering during the outdoor experience, state connection with nature, items about the characteristics of the site visited, and the perceived restorativeness of the nature experience (perceived restorativeness is not reported on in this study).

2.4. Measures

2.4.1. Trait and state mindfulness

Trait mindfulness was measured with the 14-item version of the Freiburg Mindfulness Inventory (FMI-14; Walach et al., 2006). The FMI-14 was developed for the general population, and uses items that measure tendency to be present, such as "*I am open to the experience of the present moment*", and to have an accepting attitude: "*I accept unpleasant experiences*". The Likert-response scale ranges from 1 (rarely) to 4 (almost always). One item was reverse-coded and scores were summed, resulting in good internal consistency (Cronbach's $\alpha = .81$; Cronbach,

Table 1
Demographic characteristics of participants in full sample and each intervention group.

	Mindful engagement $n = 49$		Directed engagement $n = 55$		Mind wandering $n = 57$		Unguided control $n = 54$		Full sample $N = 215$		χ^2 or ANOVA testing group differences
	n	%	n	%	n	%	n	%	n	%	
Gender											$\chi^2(2, 3) = 7.65, p = .57$
Female	36	74	45	82	40	70	39	72	160	74	
Male	13	27	10	18	16	28	14	26	53	25	
Non-binary	0	0	0	0	1	2	0	0	1	<1	
Highest education level											$\chi^2(3, 3) = 15.27, p = .08$
Year 12 or equivalent	24	49	19	35	25	44	29	54	97	45	
Vocational qualification	1	2	0	0	1	2	0	0	2	1	
Undergraduate degree	15	31	23	42	17	30	23	43	78	36	
Postgraduate degree	9	18	13	24	14	25	2	3.7	38	18	
Employment											$\chi^2(2, 3) = 3.78, p = .71$
Student	38	77	40	73	40	70	41	76	156	73	
Employed	11	22	14	26	17	30	13	24	58	27	
Unemployed	0	0	1	<1	0	0	0	0	1	<1	
Country of current residence											$\chi^2(2, 3) = 9.59, p = .14$
Australia	39	80	31	56	42	74	38	70	150	70	
China	6	8	16	29	7	12	8	13	37	16	
Other	4	12	8	15	8	14	8	17	28	14	
Age	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	Mean	*SD*	
	24.1	8.9	24.7	9.5	26.3	11.1	22.3	7.3	24.3	9.4	$F(3, 210) = 1.69, p = .17$

1951).

State mindfulness was measured with items from three validated measures: the State Mindfulness Scale (SMS; Tanay & Bernstein, 2013), the Multidimensional State Mindfulness Questionnaire (MSMQ; Blanke & Brose, 2017), and the Toronto Mindfulness Scale (TMS; Lau et al., 2006). The measures were chosen to reflect the multidimensional qualities of state mindfulness (Blanke & Brose, 2017) and the theorised mechanisms underpinning mindful awareness in nature experiences (Macaulay et al., 2022). The SMS-12 measures the mental and physical qualities of mindfulness (Cox et al., 2016), and includes items such as *"I noticed emotions come and go"*. The decentering subscale of the TMS was used, and includes items such as *"I was aware of my thoughts and feelings without overidentifying with them"*. MSMQ subscales of nonjudgmental acceptance and nonreactivity were included, using items such as *"I judged the things that were running through my head"*. In total, state mindfulness was measured with 26 items. Participants were asked to respond how well each statement reflected their outdoor experience, with responses ranging from 1 (Not at all) to 5 (Extremely). The nonjudgmental acceptance items were reverse-coded, and scores were summed: higher scores indicate a higher level of state mindfulness (α = .91).

2.4.2. Connection with nature

The connectedness with nature scale (CNS) measures a sense of oneness with nature (Mayer & Frantz, 2004). The CNS-7 (Pasca et al., 2017) was used to measure both pre-test trait-level connection with nature (for example, *"I often feel disconnected from nature"*; α = .84), and post-test state-level connection with nature (items re-worded to suit the current state, for example, *"At the present moment, I don't feel connected to nature"*; α = .90). Scores were summed and higher scores indicate a person feels more connected with nature.

2.4.3. Positive and negative affect

To measure mood, we used the positive affect (PA) and negative affect (NA) scales (PANAS; Watson et al., 1988). We asked participants to indicate the extent that they felt different feelings and emotions '*right now, that is, at the present moment*'. Responses range from 1 (very slightly or not at all) to 5 (extremely), and were summed as separate scales (post-test PA α = .91; post-test NA α = .90).

2.4.4. Sustained attention

The random version of the Sustained Attention to Response Task (SART; Johnson et al., 2007; Robertson et al., 1997Johnson et al., 2007; Robertson et al., 1997) was administered online using Pavlovia (https://pavlovia.org/). In the task, a random sequence of single digits was presented on the screen. Participants were instructed to respond quickly and accurately by pressing the spacebar of the keyboard to all digits except '3'. Each digit was shown for 250 ms followed by a white cross in a circle, for 900 ms. The interstimulus interval was 1150 ms, there were 225 trials, and each SART lasted 4m 30s. Both the pre-test and post-test tasks started with a 10-digit practice trial.

The mean and standard deviation (SD) response time (RT) were measured for each digit except '3'. Omission errors were counts of failures to respond to the go digits, and commission errors were counts of incorrect responses to the no-go '3'. Participants were excluded if they scored >30 omission errors, indicating that they had not performed the task correctly (pre-test n = 5, post-test n = 9). Moment-to-moment variability (fast frequency area under the spectra) and gradual variability (slow frequency area under the spectra) were calculated using the methodology outlined in Johnson et al. (2020). Moment-to-moment variability is a measure of all variability of RT within a sequence of 9 digits, a SART cycle. Gradual variability is a measure of all variability longer than one cycle. Higher values on these latter two variables indicate higher response variability. Each SART measure was calculated for the first (trial 2 to 113) and second (trial 114 to 225) halves of the task, to measure possible time-on-task effects and decline in vigilance over the 4.5min task. The half-by-half analysis of the SART has been useful in similar nature-based studies that find differences in performance over the first and second halves of the task (Lee et al., 2015; Pasanen et al., 2018).

2.4.5. Self-reported environmental characteristics

After the outdoor experience, participants were asked to report on the features of the outdoor environment they visited. The items were created to reflect restorative features identified in previous studies (e.g., Lindal & Hartig, 2015; Mesimäki et al., 2019; Nordh & Østby, 2013). We used this measure to ensure there was no significant variation between intervention groups on the type of environment visited. We confirmed this with a one-way ANOVA: $F(1, 211) = 1.13$, $p = .34$, $\eta^2 = .02$. The full measure and item means and standard deviations are found in Appendix C.

2.5. Analysis

All analyses were conducted using SPSS version 27. Prior to the main analyses we checked and confirmed that no groups differed on any of the pre-test measures, using one-way ANOVAs.

We compared the differences in means between the four intervention groups on psychological outcomes with ANCOVAs, while controlling for the pre-test, or trait-level scores. Assumptions of the statistical models were assessed before proceeding, including homogeneity of regression slopes, homogeneity of variance assessed by inspecting the distribution of standardized residuals and Levene's test. Cases appearing as outliers on residuals plots were removed and the models rerun as a sensitivity check. We transformed three variables (pre- and post-test) due to violating assumptions of homogeneity of variance: we used an inverse transformation for negative affect, and logarithmic transformations for both SART moment-to-moment variability and gradual variability. Omission errors were not included as a dependent variable in the analysis due to significant floor effects: 40% of participants made 0 omission errors in the pre-test attention task. Descriptive statistics of each dependent variable are presented in Table 2.

A series of mediation analyses were conducted using PROCESS in SPSS. The mediation models used categorical predictors (intervention group), and indicator coding with the unguided control group as the reference group. State mindfulness was the mediator variable, and various outcome variables were tested: connection with nature, positive affect, negative affect, and SART variables. Within each analysis, mediation was analysed on three separate levels of the predictor, indicating comparisons between engagement interventions and the reference group: the indirect effect was therefore interpreted as the impact each engagement intervention (mindful engagement, directed engagement, and mind wandering) had on the outcome variable as a result of its impact on state mindfulness, relative to the unguided control group (Fig. 2; Hayes & Preacher, 2014).

3. Results

3.1. Adherence to protocol

Participants were included in the final analyses if they completed the outdoor intervention in an acceptable time, between 20 and 45 min (n = 215). There was no statistical difference between groups for the amount of time spent during the intervention, $F(3, 311) = 0.13$, $p = .94$.

3.2. Attention fatigue

Paired samples t-tests indicated that participants' sustained attention became fatigued during the first SART. Omission errors, $t(194) = -4.26$, $p < .001$, SDRT, $t(194) = -3.03$, $p = .003$, moment-to-moment variability, $t(192) = -3.82$, $p < .001$, and gradual variability, $t(192) = -4.91$, $p < .001$, increased from the first to the second half of the first

Table 2
Descriptive Statistics: Means (M) and Standard Deviations (SD) of Pre- and Post-test Measures for each Experimental Group.

Measure	Mindful engagement		Directed engagement		Mind wandering		Unguided control	
	Pre-test *M (SD)*	Post-test *M (SD)*	Pre-test *M (SD)*	Post-test *M (SD)*	Pre-test *M (SD)*	Post-test *M (SD)*	Pre-test *M (SD)*	Post-test *M (SD)*
Mindfulness[a]	36.5 (5.7)	86.3 (14.4)	33.7 (6.6)	81.0 (15.4)	34.5 (6.2)	81.6 (15.0)	33.7 (5.5)	76.1 (14.4)
Connection with nature	25.8 (4.8)	26.9 (5.6)	23.4 (5.0)	24.9 (4.7)	24.5 (4.5)	25.3 (4.7)	24.9 (4.6)	25.6 (4.9)
Positive affect	26.7 (6.6)	27.1 (8.4)	25.8 (7.0)	25.6 (8.2)	26.4 (7.2)	26.2 (7.9)	25.1 (6.7)	25.2 (7.8)
Negative affect	17.5 (7.2)	14.0 (4.7)	18.8 (7.8)	15.8 (5.6)	18.2 (6.8)	13.4 (3.9)	18.9 (8.5)	15.6 (7.8)
SART Mean RT (FH)	411 (86)	424 (89)	417 (108)	412 (104)	406 (92)	411 (110)	407 (87)	412 (103)
SART Mean RT (SH)	407 (91)	397 (94)	420 (123)	401 (104)	390 (104)	387 (92)	396 (109)	402 (107)
SART SDRT (FH)	95 (32)	94 (44)	88 (34)	98 (50)	88 (42)	88 (41)	89 (33)	77 (29)
SART SDRT (SH)	106 (42)	106 (47)	101 (50)	104 (56)	92 (49)	87 (39)	88 (35)	84 (37)
SART commission errors (FH)	5.8 (3.5)	5.0 (2.9)	6.0 (4.0)	6.3 (4.1)	6.1 (4.0)	5.8 (4.6)	6.0 (4.0)	4.7 (3.9)
SART commission errors (SH)	3.9 (2.2)	4.0 (2.3)	4.0 (2.3)	3.8 (2.4)	4.0 (2.5)	3.5 (2.6)	3.8 (2.2)	3.1 (2.6)
SART moment-to-moment variability (FH)	92 (72)	89 (75)	70 (44)	98 (94)	87 (117)	67 (54)	84 (58)	62 (52)
SART moment-to-moment variability (SH)	134 (103)	112 (98)	119 (134)	121 (128)	111 (153)	88 (67)	90 (78)	91 (84)
SART gradual variability (FH)	376 (345)	413 (529)	330 (449)	506 (545)	373 (675)	309 (287)	387 (530)	289 (503)
SART gradual variability (SH)	540 (492)	545 (453)	599 (706)	510 (574)	475 (606)	469 (779)	401 (331)	404 (495)

All variables are untransformed. FH = first half; SH = second half.
[a] Pre- and post-test measures are different: trait mindfulness and state mindfulness.

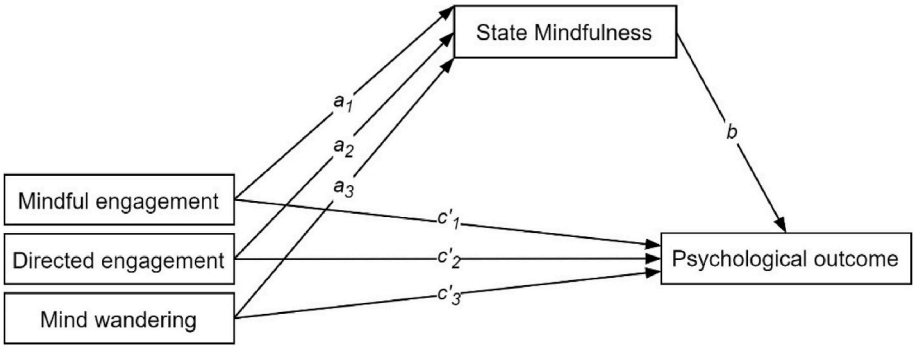


Fig. 2. Conceptual representation of the analysed mediation pathways. Predictor variable is multi-categorical with >2 levels (four interventions groups): indirect effects are thus examined separately with the unguided control group as the reference category (a_1b, a_2b, a_3b).

SART. Mean response time remained consistent across the two halves of the first SART, $t(194) = 1.62$, $p = .10$, and commission errors reduced in the second half of the task, $t(194) = 9.04$, $p < .001$.

3.3. Engagement intervention group differences on psychological outcomes

3.3.1. State mindfulness

There was a significant difference between intervention groups for state mindfulness experienced in nature, as reported in the post-test survey, while controlling for trait-level mindfulness, $F(3, 198) = 2.90$, $p = .04$, partial $\eta^2 = .04$. Pairwise comparisons revealed that the mindful engagement group had a higher level of experienced state mindfulness than the unguided control group ($p < .01$), but no significant difference compared with other groups (Fig. 3a).

3.3.2. Connection with nature

After the nature experience, there was no difference between intervention groups for state-level connection with nature, while controlling for the trait-level connection with nature, $F(3, 202) = 0.46$, $p = .71$, partial $\eta^2 = .01$ (Fig. 3b).

3.3.3. Positive affect

There was no difference between intervention groups for positive affect after the nature experience, while controlling for pre-test positive affect, $F(3, 206) = 0.34$, $p = .79$, partial $\eta^2 = .01$ (Fig. 3c).

3.3.4. Negative affect

After the nature experience, there was a significant difference between intervention groups for negative affect, while controlling for the pre-test negative affect, $F(3, 204) = 2.99$, $p = .03$, partial $\eta^2 = .04$. Pairwise comparisons revealed that the directed engagement group had significantly higher negative affect than the mind wandering group ($p < .01$), but there were no significant differences between other groups (Fig. 3d).

3.3.5. SART mean RT

There was no difference between intervention groups for the post-test, first half mean RT when controlling for the pre-test first half measure, $F(3, 174) = 1.58$, $p = .20$, partial $\eta^2 = 0.03$, nor for the second half mean RT, $F(3, 173) = 0.99$, $p = .40$, partial $\eta^2 = 0.02$ (Fig. 4a).

3.3.6. SART SDRT

There was a significant effect on post-test first half SDRT on levels of intervention group when controlling for the pre-test first half measure, $F(3, 175) = 4.60$, $p < .01$, partial $\eta^2 = 0.07$. Pairwise comparisons revealed that the unguided control was significantly less variable in responding than the directed engagement group ($p < .001$), the mindful engagement group ($p = .01$) and the mind wandering group ($p = .01$). There was no significant effect on post-test second half SDRT on levels of intervention group when controlling for the pre-test second half score, $F(3, 175) = 1.57$, $p = .20$, partial $\eta^2 = 0.03$ (Fig. 4b).

3.3.7. SART commission errors

After the nature experience, there was a significant difference

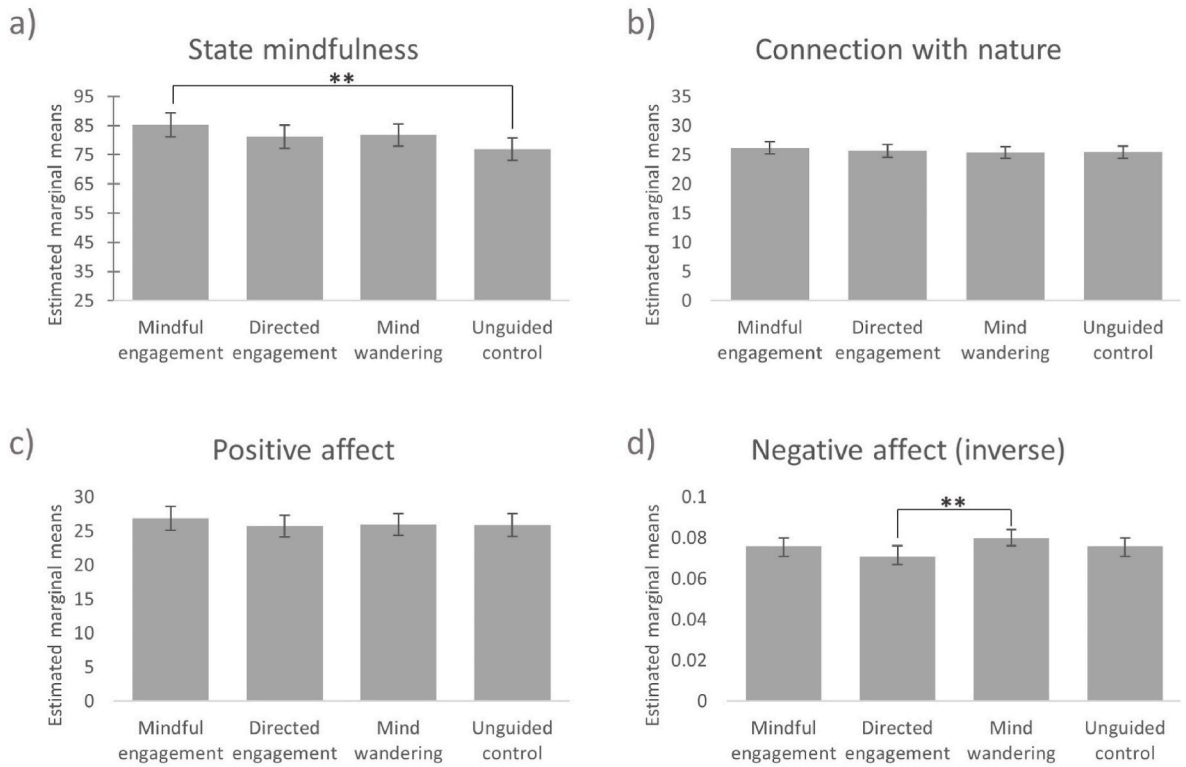


Fig. 3. Estimated marginal means in different intervention groups for the self-reported post-test measures, controlling for pre-test baseline scores.
** Indicates significant differences between groups, $p < .01$.
Higher score here on negative affect (inverse score) represents lower negative affect.

between intervention groups for first-half commission errors, while controlling for the pre-test first half scores, $F(3, 175) = 2.88$, $p = .04$, partial $\eta^2 = .05$). Pairwise comparisons revealed that the directed engagement group made more commission errors than the unguided control group ($p = .04$), and the mindful engagement group ($p = .02$). There was no difference between intervention groups for second-half commission errors, while controlling for the pre-test second half scores, $F(3, 175) = 1.89$, $p = .13$, partial $\eta^2 = .03$ (Fig. 4c).

3.3.8. SART gradual variability

There was a significant difference between intervention groups for post-test first half gradual variability, while controlling for the pre-test first half scores, $F(3, 169) = 5.85$, $p < .01$, partial $\eta^2 = .09$). Pairwise comparisons revealed that the directed engagement group was significantly more variable in responding than the unguided control group ($p < .001$), the mindful engagement group ($p = .02$), and the mind wandering group ($p = .01$). There was no difference between intervention groups for second-half gradual variability, while controlling for the pre-test second-half scores, $F(3, 170) = 1.78$, $p = .15$, partial $\eta^2 = .03$, (Fig. 4d).

3.3.9. SART moment-to-moment variability

After the nature experience, there was a significant difference between intervention groups for first half moment-to-moment variability, while controlling for the pre-test first half scores, $F(3, 169) = 4.09$, $p = .01$, partial $\eta^2 = .07$. Pairwise comparisons revealed that the directed engagement group was more variable in responding than the unguided control group ($p = .01$), and the mind wandering group ($p = .03$). The unguided control group had the lowest variability overall, and was also significantly lower than the mindful engagement group ($p = .04$). There was no difference between intervention groups for second half moment-to-moment variability, while controlling for the pre-test second half scores, $F(3, 170) = 0.75$, $p = .53$, partial $\eta^2 = .01$ (Fig. 4e).

3.4. Indirect effects of engagement interventions on psychological outcomes through state mindfulness

3.4.1. Connection with nature

Relative to the unguided control group, all three engagement interventions indirectly influenced connection with nature through a higher level of experienced state mindfulness, where all bootstrapped confidence intervals for the indirect effects were above zero: mindful engagement indirect effect = 0.81, 95% CI = 0.32, 1.44, directed engagement indirect effect = 0.53, 95% CI = 0.05, 1.12, mind wandering indirect effect = 0.52, 95% CI = 0.05, 1.07. Relative to the unguided control group, there was no evidence that the engagement interventions led to increases in connection with nature independently of their effect on state mindfulness (direct effects, Fig. 5a).

3.4.2. Positive affect and negative affect

Relative to the unguided control group, the mindful engagement intervention indirectly influenced both positive affect and negative affect through the higher level of experienced state mindfulness, where the bootstrapped confidence interval was above zero: indirect effect on positive affect = 1.44, 95% CI = 0.56, 2.50, indirect effect on negative affect = −0.38, 95% CI = −0.86, −0.04. The directed engagement and mind wandering interventions did not indirectly influence positive affect or negative affect relative to the unguided control group. Relative to the unguided control group, there was no evidence that the engagement interventions led to increases in positive affect (direct effects, Fig. 5b) or negative affect (direct effects, Fig. 5c) independently of their effect on state mindfulness.

3.4.3. SART mean RT and commission errors

Relative to the unguided control group, the mindful engagement intervention indirectly influenced mean RT and number of commission errors in the first half of the attention task through the higher level of

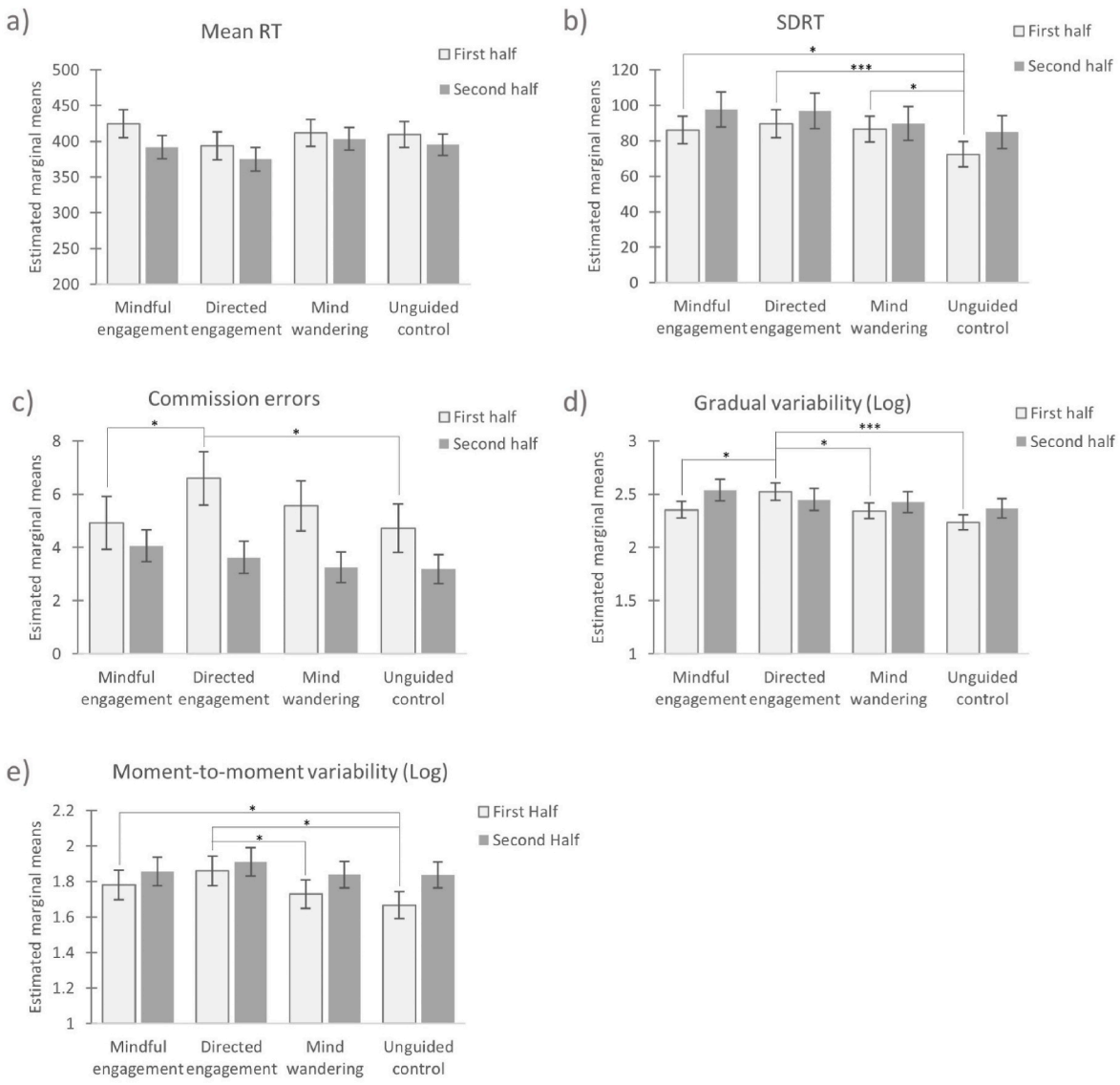


Fig. 4. Estimated marginal means in different intervention groups for the post-test attention scores, controlling for pre-test baseline scores. Error bars show 95% confidence intervals.
* Indicates significant differences between groups, $p < .05$; ***$p < .001$.

experienced state mindfulness: indirect effect on mean RT = 7.85, *SE* = 3.69, 95% CI = 1.62, 16.09 (Fig. 5e), indirect effect on number of commission errors = −0.46, 95% CI = −0.98, −0.11 (Fig. 5d). There were no direct effects of the engagement interventions on mean RT. There was a direct effect of the directed engagement intervention on commission errors in the first half, where the directed engagement intervention led to higher number of commission errors relative to the unguided control group, and independent of its effect on state mindfulness. There was no evidence of an indirect effect or direct effect on mean RT or commission errors in the second half of the attention task (Appendix B).

3.4.4. SART SDRT

There was no evidence of an indirect effect on SDRT in the first half of the attention task (Appendix B). In the second half of the task, the mindful engagement intervention indirectly influenced SDRT relative to the unguided control group, through the higher level of experienced state mindfulness: mindful engagement indirect effect −3.59, *SE* = 2.06, 95% CI = −8.17, −0.28 (Fig. 5f).

3.4.5. SART moment-to-moment and gradual variability

There was no evidence of the engagement interventions indirectly influencing gradual variability or moment-to-moment variability on the attention task in the first or second half (Appendix B).

4. Discussion

The present study compared the efficacy of three brief engagement interventions in nature on psychological outcomes, revealing that different types of engagement in nature may be suited to distinct outcomes. Our findings indicated that the mindfulness intervention led to higher levels of state mindfulness than the unguided control group, though not a statistically higher level than the directed engagement and mind wandering groups (partially supporting H1a). Although it is possible that the mindful engagement gained a better understanding of mindfulness through the intervention tasks, and therefore responded differently on the state mindfulness questionnaire (Van Dam et al., 2012), the lack of difference between the other groups suggested that the contrast with the unguided control group was meaningful. It was expected that the mindful engagement group would have significantly

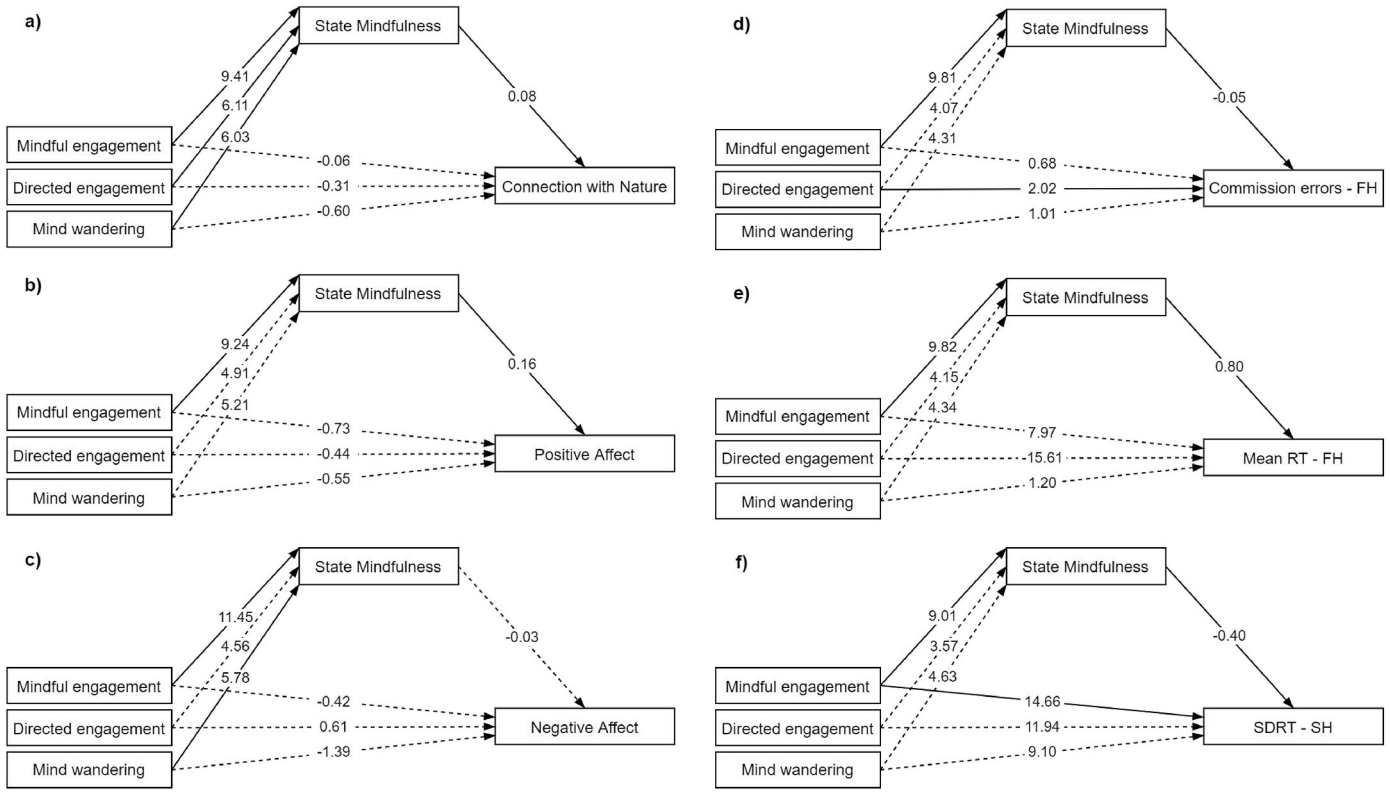


Fig. 5. Mediation models representing the effect of the intervention groups (mindful engagement, directed engagement, and mind wandering) on connection with nature (a), positive affect (b), negative affect (c), commission errors first half (d), mean reaction time second half (e), and standard deviation reaction time (f). Models include the direct effect and the mediating effect of state mindfulness experienced in nature, as reported in the post-test survey. Significant pathways are solid lines. All models used indicator coding with the unguided control group as the reference group. FH = First Half; SH = Second Half.

higher levels of state mindfulness than the directed engagement and mind wandering groups, consistent with previous research that compares mindfulness with other engagement interventions (Aspy & Proeve, 2017; Rahl et al., 2017) including mind wandering (Arch & Craske, 2006; Kiken & Shook, 2011). Other studies that use brief (15–20min) mindfulness interventions in nature have similarly observed no significant effects on state mindfulness (Nisbet et al., 2019; Unsworth et al., 2016).

There were significant differences between the intervention groups on the following outcomes: negative affect, and four measures on the first half of the post-test attention task – commission errors, SDRT, moment-to-moment variability, and gradual variability. The differences between groups were not in the directions hypothesised (H1 and H2). The directed engagement group experienced significantly higher negative affect than the mind wandering group. Although the directed engagement group were asked to spend the final 5 min noticing beauty in nature, which can lead to positive wellbeing outcomes (Capaldi et al., 2017), they were also asked to be reflective and judge their environment, which may have led to negative emotional responses (Pasanen et al., 2018). The finding that the mind wandering group had the lowest levels of negative affect was not expected, but perhaps unsurprising, given the mixed literature relating to the possible benefits of mind wandering in the right context (Schooler et al., 2014). Franklin et al. (2013) found that although instances of mind wandering are typically associated with lower mood, the opposite is true when thought content is interesting and useful. We propose that being in nature and on a work break, may provide an interesting context for mind wandering, and explain our finding that this group had the best mood outcomes. This argument is supported by a recent study that examined the contents of written thoughts during outdoor experiences, and found that these experiences are commonly associated with positive introspection (Schertz et al., 2018).

Performance on the post-test attention task demonstrated that the unguided control group had the highest level of attention restoration during the nature experience, and that the directed engagement group had the lowest level of attention restoration. For the first half of the post-test attention task, the unguided control group performed significantly better on each of the measures (lower commission errors, SDRT, moment-to-moment and gradual variability) than the directed engagement group, and performed better than all groups on SDRT. The directed engagement group had lower performance than all groups on the first half of the attention task for gradual variability, and lower performance on commission errors than the mindful engagement group. By the second half of the task, all groups were performing similarly. We offer three possible explanations for these findings. First, it is plausible that the exercises were somewhat cognitively effortful for participants, resulting in poorer attention performance for the directed engagement group and the highest attention restoration for the unguided group. Lymeus et al. (2017) demonstrate that mindful engagement exercises incur some initial attentional effort, and this may also be the case for our engagement exercises. Second, the unguided control group did not have to use their phone during the outdoor experience: previous research shows that engaging with technology during an outdoor break is detrimental to attention restoration (Jiang et al., 2019). Third, as noted above, the directed engagement group's intervention exercises may have directly contributed to a less restorative experience for this group. These findings overall raise the question of how the unguided control group engaged during their experience. Although we did not instruct or assess their activity in the environment, the context of a work break may have allowed this group to engage in activities that suited their needs and, as the results suggest, avoid tasks that tax attention.

A surprising finding was that there was no difference between intervention groups on connection with nature after the nature experience. Similar studies have found that higher engagement during short

nature experiences lead to a stronger sense of nature connection, with large effects (Lumber et al., 2017; Nisbet et al., 2019). We note that Lumber et al.'s engagement intervention was specifically designed to strengthen nature connection based on theoretically-grounded pathways to this outcome. While the mindful engagement and directed engagement interventions in the current study involved some engagement to the external, natural environment, the interventions also involved other concepts such as observing body and thoughts. Interventions that are designed to target specific outcomes from engagement in nature may be most beneficial (for example, Lumber et al.'s nature connection exercises (2017), or Pasanen et al.'s restoration-enhancement exercises (2018)).

There is partial support for our hypothesis that mindfulness during the experience would function as a mediator of the effect of the engagement intervention on psychological outcomes. For all three intervention groups compared with the unguided control group, the engagement intervention resulted in a higher level of connection with nature, as a result of the positive effect of the intervention on experienced state mindfulness. For positive affect, negative affect, mean RT, SDRT, and commission errors, indirect effects through state mindfulness were observed only for the mindful engagement group. These findings are consistent with past research that links state mindfulness in nature with connection with nature (Nisbet et al., 2019) and affect (Stewart & Haaga, 2018). Other studies have shown that simply being in nature increases state mindfulness (Brymer et al., 2021; Hamann & Ivtzan, 2016). Stewart and Haaga (2018) demonstrated that state mindfulness may mediate the effect of exposure to virtual nature on positive affect and psychological wellbeing. Our findings are consistent with these studies, though we extend this work by proposing an indirect pathway between specific engagement techniques and these outcomes as a result of state mindfulness experienced in nature.

Several limitations of the study should be noted. We designed the intervention to be entirely online, which posed both strengths and weaknesses to the overall study. A strength of this design was that the intervention could be delivered to individuals under a range of circumstances, including restrictions associated with the COVID-19 pandemic. Weaknesses, however, included a high rate of attrition, being unable to monitor compliance with intervention instructions, and the use of smartphones during the outdoor break – which may have been disruptive to restoration (Jiang et al., 2019). Further to this, participants could spend their break how they preferred, as they were not given instructions other than the engagement exercises; although this design feature improves ecological validity, there was likely much variation in the experience of different participants that was not monitored. Future research may explore similar interventions that deliver instructions in-person and/or use a measure that assesses individual behaviour and activities during the experience. The nature of the online intervention also meant that participants could choose any outdoor environment near their home, so some environments may not have had many natural features at all. This limitation may have impacted the mindful and directed engagement groups who were asked to notice and find beauty in nature; however, even in the most constrained outdoor environments, viewing the sky, clouds and weather can function as a positive source of restoration and nature contact (T.-Y. Lin et al., 2022; Richardson et al., 2015).

4.1. Conclusion

Our study directly compared mindful and other forms of engagement in nature. It contributed to a better understanding of the role of engagement in nature experiences, showing that such interventions need to be carefully designed and applied. First, interventions designed to heighten *mindful* engagement in nature may benefit from alternative designs or applications (for example, using guided meditations, Lymeus et al., 2018), as our mindful engagement group did not experience a higher level of state mindfulness than the other two experimental groups. Second, simply engaging more with an experience does not necessarily lead to benefit: the directed engagement group had the least restorative experience in this study. Instructions must therefore be designed with specific intention. If one is looking to restore attention, for example, our results indicate that an unguided experience may be most useful. On the other hand, mind wandering in nature may improve mood greater than other forms of engagement. To our knowledge this is the first study that applied mind wandering as an experimental intervention in nature, and further research is needed to understand whether and how mind wandering may benefit different types of nature experiences.

Author statement

Rose Macaulay: Conceptualization, Data Collection, Formal Analysis, Writing- Original draft preparation. **Kate Lee**: Conceptualization, Supervision, Writing- Reviewing and Editing. **Katherine Johnson**: Conceptualization, Supervision, Writing- Reviewing and Editing. **Kathryn Williams**: Conceptualization, Supervision, Writing- Reviewing and Editing.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvp.2022.101813.

References

Arch, J. J., & Craske, M. G. (2006). Mechanisms of mindfulness: emotion regulation following a focused breathing induction. *Behaviour Research and Therapy, 44*(12), 1849–1858. https://doi.org/10.1016/j.brat.2005.12.007

Aspy, D. J., & Proeve, M. (2017). Mindfulness and loving-kindness meditation: Effects on connectedness to humanity and to the natural world. *Psychological Reports, 120*(1), 102–117. https://doi.org/10.1177/0033294116685867

Bishop, S. R., Lau, M., Shapiro, S., Carlson, L., Anderson, N. D., Carmody, J., Segal, Z. V., Abbey, S., Speca, M., Velting, D., & Devins, G. (2006). Mindfulness: A proposed operational definition. *Clinical Psychology: Science and Practice, 11*(3), 230–241. https://doi.org/10.1093/clipsy.bph077

Blanke, E. S., & Brose, A. (2017). Mindfulness in daily life: A multidimensional approach. *Mindfulness, 8*(3), 737–750. https://doi.org/10.1007/s12671-016-0651-4

Brown, D. K., Barton, J. L., Pretty, J., & Gladwell, V. F. (2012). Walks4work: Rationale and study design to investigate walking at lunchtime in the workplace setting. *BMC Public Health, 12*(1), 550. https://doi.org/10.1186/1471-2458-12-550

Brymer, E., Crabtree, J., & King, R. (2021). Exploring perceptions of how nature recreation benefits mental wellbeing: A qualitative enquiry. *Annals of Leisure Research, 24*(3), 394–413. https://doi.org/10.1080/11745398.2020.1778494

Capaldi, C. A., Passmore, H.-A., Ishii, R., Chistopolskaya, K. A., Vowinckel, J., Nikolaev, E. L., & Semikin, G. I. (2017). Engaging with natural beauty may Be related to well-being because it connects people to nature: evidence from three cultures. *Ecopsychology, 9*(4), 199–211. https://doi.org/10.1089/eco.2017.0008

Cardaciotto, L., Herbert, J. D., Forman, E. M., Moitra, E., & Farrow, V. (2008). The assessment of present-moment awareness and acceptance: the philadelphia mindfulness scale. *Assessment, 15*(2), 204–223. https://doi.org/10.1177/1073191107311467

Cervinka, R., Schwab, M., & Haluza, D. (2020). Investigating the qualities of a recreational forest: findings from the cross-sectional hallerwald case study. *International Journal of Environmental Research and Public Health, 17*(5), 1676. https://doi.org/10.3390/ijerph17051676

Choe, E. Y., Jorgensen, A., & Sheffield, D. (2020). Does a natural environment enhance the effectiveness of mindfulness-based stress reduction (MBSR)? Examining the mental health and wellbeing, and nature connectedness benefits. *Landscape and Urban Planning, 202*, 103886. https://doi.org/10.1016/j.landurbplan.2020.103886

Christoff, K., Irving, Z. C., Fox, K. C. R., Spreng, R. N., & Andrews-Hanna, J. R. (2016). Mind-wandering as spontaneous thought: a dynamic framework. *Nature Reviews Neuroscience, 17*(11), 718–731. https://doi.org/10.1038/nrn.2016.113

Cox, A. E., Ullrich-French, S., & French, B. F. (2016). Validity evidence for the state mindfulness scale for physical activity. *Measurement in Physical Education and Exercise Science, 20*(1), 38–49. https://doi.org/10.1080/1091367X.2015.1089404

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika, 16*(3), 297–334. https://doi.org/10.1007/BF02310555

Duvall, J. (2011). Enhancing the benefits of outdoor walking with cognitive engagement strategies. *Journal of Environmental Psychology, 31*(1), 27–35. https://doi.org/10.1016/j.jenvp.2010.09.003

Faul, F., Erdfelder, E., Lang, A.-G., & Buchner, A. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods, 39*(2), 175–191. https://doi.org/10.3758/BF03193146

Franklin, M. S., Mrazek, M. D., Anderson, C. L., Smallwood, J., Kingstone, A., & Schooler, J. W. (2013). The silver lining of a mind in the clouds: interesting musings

are associated with positive mood while mind-wandering. *Frontiers in Psychology, 4*. https://doi.org/10.3389/fpsyg.2013.00583
Fritz, M. S., & Mackinnon, D. P. (2007). Required sample size to detect the mediated effect. *Psychological Science, 18*(3), 233–239. https://doi.org/10.1111/j.1467-9280.2007.01882.x. PubMed.
Hamann, G. A., & Ivtzan, I. (2016). 30 minutes in nature a day can increase mood, well-being, meaning in life and mindfulness: effects of a pilot programme. *Social Inquiry Into Well-Being, 2*(2), 34–46.
Hayes, A. F., & Preacher, K. J. (2014). Statistical mediation analysis with a multicategorical independent variable. *British Journal of Mathematical and Statistical Psychology, 67*(3), 451–470. https://doi.org/10.1111/bmsp.12028
Jiang, B., Schmillen, R., & Sullivan, W. C. (2019). How to waste a break: Using portable electronic devices substantially counteracts attention enhancement effects of green spaces. *Environment and Behavior, 51*(9–10), 1133–1160. https://doi.org/10.1177/0013916518788603
Johnson, K. A., Robertson, I. H., Kelly, S. P., Silk, T. J., Barry, E., Dáibhis, A., Watchorn, A., Keavey, M., Fitzgerald, M., Gallagher, L., Gill, M., & Bellgrove, M. A. (2007). Dissociation in performance of children with ADHD and high-functioning autism on a task of sustained attention. *Neuropsychologia, 45*(10), 2234–2245. https://doi.org/10.1016/j.neuropsychologia.2007.02.019
Johnson, K. A., White, M., Wong, P. S., & Murrihy, C. (2020). Aspects of attention and inhibitory control are associated with on-task classroom behaviour and behavioural assessments, by both teachers and parents, in children with high and low symptoms of ADHD. *Child Neuropsychology, 26*(2), 219–241. https://doi.org/10.1080/09297049.2019.1639654
Kabat-Zinn, J. (2003). Mindfulness-based interventions in context: past, present, and future. *Clinical Psychology: Science and Practice, 10*, 144–156. https://doi.org/10.1093/clipsy.bpg016
Kaplan, S. (1995). The restorative benefits of nature: Toward an integrative framework. *Journal of Environmental Psychology, 15*(3), 169–182. https://doi.org/10.1016/0272-4944(95)90001-2
Kaplan, S. (2001). Meditation, restoration, and the management of mental fatigue. *Environment and Behavior, 33*(4), 480–506. https://doi.org/10.1177/00139160121973106
Kiken, L. G., & Shook, N. J. (2011). Looking up: Mindfulness increases positive judgments and reduces negativity bias. *Social Psychological and Personality Science, 2*(4), 425–431. https://doi.org/10.1177/1948550610396585
Killingsworth, M. A., & Gilbert, D. T. (2010). A wandering mind is an unhappy mind. *Science, 330*(6006). https://doi.org/10.1126/science.1192439, 932–932.
Korpela, K., Savonen, E.-M., Anttila, S., Pasanen, T., & Ratcliffe, E. (2017). Enhancing wellbeing with psychological tasks along forest trails. *Urban Forestry and Urban Greening, 26*, 25–30. https://doi.org/10.1016/j.ufug.2017.06.004
Lau, M. A., Bishop, S. R., Segal, Z. V., Buis, T., Anderson, N. D., Carlson, L., Shapiro, S., Carmody, J., Abbey, S., & Devins, G. (2006). The Toronto mindfulness scale: Development and validation. *Journal of Clinical Psychology, 62*(12), 1445–1467. https://doi.org/10.1002/jclp.20326
Lee, K. E., Williams, K. J. H., Sargent, L. D., Williams, N. S. G., & Johnson, K. A. (2015). 40-second green roof views sustain attention: the role of micro-breaks in attention restoration. *Journal of Environmental Psychology, 42*, 182–189. https://doi.org/10.1016/j.jenvp.2015.04.003
Lindal, P. J., & Hartig, T. (2015). Effects of urban street vegetation on judgments of restoration likelihood. *Urban Forestry and Urban Greening, 14*(2), 200–209. https://doi.org/10.1016/j.ufug.2015.02.001
Lin, T.-Y., Le, A.-V., & Chan, Y.-C. (2022). Evaluation of window view preference using quantitative and qualitative factors of window view content. *Building and Environment, 213*, 108886. https://doi.org/10.1016/j.buildenv.2022.108886
Lin, Y.-H., Tsai, C.-C., Sullivan, W. C., Chang, P.-J., & Chang, C.-Y. (2014). Does awareness effect the restorative function and perception of street trees? *Frontiers in Psychology, 5*. https://doi.org/10.3389/fpsyg.2014.00906
Lumber, R., Richardson, M., & Sheffield, D. (2017). Beyond knowing nature: Contact, emotion, compassion, meaning, and beauty are pathways to nature connection. *PLoS One, 12*(5), Article e0177186. https://doi.org/10.1371/journal.pone.0177186
Lymeus, F., Lindberg, P., & Hartig, T. (2018). Building mindfulness bottom-up: meditation in natural settings supports open monitoring and attention restoration. *Consciousness and Cognition, 59*, 40–56. https://doi.org/10.1016/j.concog.2018.01.008
Lymeus, F., Lundgren, T., & Hartig, T. (2017). Attentional effort of beginning mindfulness training is offset with practice directed toward images of natural scenery. *Environment and Behavior, 49*(5), 536–559. https://doi.org/10.1177/0013916516657390
Macaulay, R., Lee, K., Johnson, K., & Williams, K. (2022). Mindful engagement, psychological restoration, and connection with nature in constrained nature experiences. *Landscape and Urban Planning, 217*(104263). https://doi.org/10.1016/j.landurbplan.2021.104263
MacCoon, D. G., Imel, Z. E., Rosenkranz, M. A., Sheftel, J. G., Weng, H. Y., Sullivan, J. C., Bonus, K. A., Stoney, C. M., Salomons, T. V., Davidson, R. J., & Lutz, A. (2012). The validation of an active control intervention for Mindfulness Based Stress Reduction (MBSR). *Behaviour Research and Therapy, 50*(1), 3–12. https://doi.org/10.1016/j.brat.2011.10.011
Mayer, F. S., & Frantz, C. M. (2004). The connectedness to nature scale: A measure of individuals' feeling in community with nature. *Journal of Environmental Psychology, 24*(4), 503–515. https://doi.org/10.1016/j.jenvp.2004.10.001
McMahan, E. A., & Estes, D. (2015). The effect of contact with natural environments on positive and negative affect: a meta-analysis. *The Journal of Positive Psychology, 10*(6), 507–519. https://doi.org/10.1080/17439760.2014.994224
Mesimäki, M., Hauru, K., & Lehvävirta, S. (2019). Do small green roofs have the possibility to offer recreational and experiential benefits in a dense urban area? A case study in helsinki, Finland. *Urban Forestry and Urban Greening, 40*, 114–124. https://doi.org/10.1016/j.ufug.2018.10.005
Nisbet, E. K., Zelenski, J. M., & Grandpierre, Z. (2019). Mindfulness in nature enhances connectedness and mood. *Ecopsychology, 11*(2), 81–91. https://doi.org/10.1089/eco.2018.0061
Nordh, H., & Østby, K. (2013). Pocket parks for people – a study of park design and use. *Urban Forestry and Urban Greening, 12*(1), 12–17. https://doi.org/10.1016/j.ufug.2012.11.003
Ohly, H., White, M. P., Wheeler, B. W., Bethel, A., Ukoumunne, O. C., Nikolaou, V., & Garside, R. (2016). Attention restoration theory: A systematic review of the attention restoration potential of exposure to natural environments. *Journal of Toxicology and Environmental Health, Part B, 19*(7), 305–343. https://doi.org/10.1080/10937404.2016.1196155
Pasanen, T., Johnson, K., Lee, K., & Korpela, K. (2018). Can nature walks with psychological tasks improve mood, self-reported restoration, and sustained attention? Results from two experimental field studies. *Frontiers in Psychology, 9*, 2057. https://doi.org/10.3389/fpsyg.2018.02057
Pasca, L., Aragonés, J. I., & Coello, M. T. (2017). An analysis of the connectedness to nature scale based on item response theory. *Frontiers in Psychology, 8*, 1330. https://doi.org/10.3389/fpsyg.2017.01330
Rahl, H. A., Lindsay, E. K., Pacilio, L. E., Brown, K. W., & Creswell, J. D. (2017). Brief mindfulness meditation training reduces mind wandering: The critical role of acceptance. *Emotion, 17*(2), 224–230. https://doi.org/10.1037/emo0000250
Richardson, M., Hallam, J., & Lumber, R. (2015). One thousand good things in nature: Aspects of nearby nature associated with improved connection to nature. *Environmental Values, 24*(5), 603–619. https://doi.org/10.3197/096327115X14384223590131
Robertson, I. H., Manly, T., Andrade, J., Baddeley, B. T., & Yiend, J. (1997). Oops!': Performance correlates of everyday attentional failures in traumatic brain injured and normal subjects. *Neuropsychologia, 35*(6), 747–758. https://doi.org/10.1016/S0028-3932(97)00015-8
Schertz, K. E., Sachdeva, S., Kardan, O., Kotabe, H. P., Wolf, K. L., & Berman, M. G. (2018). A thought in the park: the influence of naturalness and low-level visual features on expressed thoughts. *Cognition, 174*, 82–93. https://doi.org/10.1016/j.cognition.2018.01.011
Schooler, J. W., Mrazek, M. D., Franklin, M. S., Baird, B., Mooneyham, B. W., Zedelius, C., & Broadway, J. M. (2014). The middle way. In *Psychology of learning and motivation, 60* pp. 1–33). Elsevier. https://doi.org/10.1016/B978-0-12-800090-8.00001-9.
Shin, Y.-K., Kim, D. J., Jung-Choi, K., Son, Y., Koo, J.-W., Min, J.-A., & Chae, J.-H. (2013). Differences of psychological effects between meditative and athletic walking in a forest and gymnasium. *Scandinavian Journal of Forest Research, 28*(1), 64–72. https://doi.org/10.1080/02827581.2012.706634
Stevenson, M. P., Schilhab, T., & Bentsen, P. (2018). Attention restoration theory II: A systematic review to clarify attention processes affected by exposure to natural environments. *Journal of Toxicology and Environmental Health, Part B, 21*(4), 227–268. https://doi.org/10.1080/10937404.2018.1505571
Stewart, M., & Haaga, D. A. F. (2018). State mindfulness as a mediator of the effects of exposure to nature on affect and psychological well-being. *Ecopsychology, 10*(1), 53–60. https://doi.org/10.1089/eco.2017.0033
Tanay, G., & Bernstein, A. (2013). State mindfulness scale (SMS): Development and initial validation. *Psychological Assessment, 25*(4), 1286–1299. https://doi.org/10.1037/a0034044
Unsworth, S., Palicki, S.-K., & Lustig, J. (2016). The impact of mindful meditation in nature on self-nature interconnectedness. *Mindfulness, 7*(5), 1052–1060. https://doi.org/10.1007/s12671-016-0542-8
Van Dam, N. T., Hobkirk, A. L., Danoff-Burg, S., & Earleywine, M. (2012). Mind your words: Positive and negative items create method effects on the five facet mindfulness questionnaire. *Assessment, 19*(2), 198–204. https://doi.org/10.1177/1073191112438743
Walach, H., Buchheld, N., Buttenmüller, V., Kleinknecht, N., & Schmidt, S. (2006). Measuring mindfulness—the Freiburg mindfulness inventory (FMI). *Personality and Individual Differences, 40*(8), 1543–1555. https://doi.org/10.1016/j.paid.2005.11.025
Watson, D., Clark, L. A., & Tellegen, A. (1988). Development and validation of brief measures of positive and negative affect: The PANAS scales. *Journal of Personality and Social Psychology, 54*(6), 1063–1070. https://doi.org/10.1037/0022-3514.54.6.1063
Weinstein, N., Przybylski, A. K., & Ryan, R. M. (2009). Can nature make us more caring? Effects of immersion in nature on intrinsic aspirations and generosity. *Personality and Social Psychology Bulletin, 35*(10), 1315–1329. https://doi.org/10.1177/0146167209341649
Williams, K. J. H., Lee, K. E., Hartig, T., Sargent, L. D., Williams, N. S. G., & Johnson, K. A. (2018). Conceptualising creativity benefits of nature experience: Attention restoration and mind wandering as complementary processes. *Journal of Environmental Psychology, 59*, 36–45. https://doi.org/10.1016/j.jenvp.2018.08.005